MD. Mehedi Hasan Maruf

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Summary

Software engineer and applied ML researcher. My B.Sc. thesis at Rajshahi University of Engineering & Technology (RUET) built TRACE, which finds the university courses that hold students back by combining prerequisite-graph analysis, Bayesian fail-rate models, and sentiment in student feedback. I then checked its answer with causal inference (DoWhy) on real student records. A paper from it is under review at LAK 2027. I also have 3+ years of industry experience building LLM and RAG features, TypeScript and Python backends, and Docker and CI/CD deployments, currently at Aamira Software Solutions.

Research

TRACE: Curriculum Bottleneck Detection & Causal Validation *Jul 2025 – Aug 2026*

B.Sc. Thesis, Dept. of CSE, RUET

- The question came from my own class. About a third of us failed one first-year mathematics course, while about 4% fail a typical course. Did that course hold students back, or only show who was already behind?
- Built TRACE to find courses like it. It ranks each course by its place in the prerequisite graph (NetworkX, GAT), a Bayesian model of its fail rate, and the tone of student feedback (DeBERTa, VADER).
- Tested how best to combine its signals on simulated curricula with planted bottlenecks (seven conditions, 30 curricula each). When each bottleneck showed up in only one signal, merging the signals' shortlists reached a top-8 precision of 0.65, against 0.32 for averaging their ranks. When bottlenecks showed up in every signal, averaging did better (0.60 against 0.41).
- Used causal inference to test what drives failure in a flagged module. On 23,369 Open University students (OULAD), a DoWhy backdoor-adjusted model estimated that high assignment engagement raised held-out pass rates by 19.9 percentage points ($p < 0.01$, E-value 1.89).
- After peer review, rebuilt the study into a paper, *When Agreement Misleads: Auditing Course Shortlists in Curriculum Analytics*, now under review at LAK 2027. It gives an exact bound on how much a combined shortlist can change top-k precision. In simulation, sources with a shared bias picked mostly the same courses (88% overlap), yet only 15% of those picks were real bottlenecks.

Experience

Software Engineer II *Jan 2025 – Present*

Aamira Software Solutions, Kokomo, Indiana, USA (remote)

- Lead three junior engineers on the company's React applications, after starting on its Angular project.
- Raised client response rates by 30% by leading development of a CRM system.
- Cut API response time by 40% by designing and building a TypeScript microservice backend.
- Made the application 60% faster by optimizing its Supabase (PostgreSQL) queries.
- Building a retrieval-augmented generation (RAG) feature for the platform.

Junior Software Engineer *Nov 2022 – Apr 2024*

Code Studio, Rajshahi, Bangladesh

- Kept 12 Linux servers at 99.8% uptime, and cut deployment time by 40% by reworking the CI/CD pipelines.
- Moved 5 client environments onto Docker, which made them simpler to deploy.
- Wrote JavaScript, TypeScript, and Python (Django) backends used by thousands of people a day.

Industrial Trainee *Mar 2025*

Brain Station 23, Dhaka, Bangladesh

- Represented a 15-person trainee cohort to management.
- Built and deployed StudentAI (see Projects), an AI learning assistant on FastAPI and Next.js.

Selected Projects

Omnious | Visual Debugging Platform *Nov 2025 – Present*

- Building a debugger that parses a TypeScript or JavaScript codebase into a graph of files, functions, and classes, then animates runtime traces over it and shows where errors cluster.
- An AI assistant suggests root causes tied to specific graph nodes, using embeddings in pgvector and a locally run LLM (Ollama). Community detection groups the graph into architectural layers.
- A TypeScript Turborepo monorepo: Fastify 5 and tRPC v11 backend, Next.js 16 / React 19 frontend, and PostgreSQL 15 on Supabase (pgvector, ltree, pg_trgm, row-level security).
- Ships a CLI indexer and Express and Next.js SDKs; runs in Docker Compose and deploys through GitHub Actions.

StudentAI | Intelligent Learning Companion *Mar 2025*

- An AI learning assistant (FastAPI, Next.js) built during industrial training at Brain Station 23.
- Answers questions about uploaded documents through a LangChain RAG pipeline with semantic retrieval and conversation memory; it also analyzes YouTube lectures and writes quizzes.
- Accepts PDFs, images, and video, and has real-time WebSocket chat, JWT login, and progress tracking.
- Deployed on Vercel and Render with a Supabase database.

Skills

AI/ML: LLM applications, RAG, LangChain, Hugging Face Transformers (DeBERTa), PyTorch, GNNs (GAT), NLP, causal inference (DoWhy), Bayesian modeling, vector search (pgvector), model evaluation (RLHF)
Languages: Python, TypeScript, JavaScript
Backend: NestJS, FastAPI, Fastify, tRPC, Django
Frontend: Next.js (React), Angular
Databases: PostgreSQL (Supabase), MongoDB, MySQL
DevOps: Docker, GitHub Actions (CI/CD), Linux server administration

Education

B.Sc. Engineering, Computer Science and Engineering *Mar 2022 – Aug 2026*

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75 / 4.00, Honours

Certifications & Achievements

- Digital Accessibility Training (DAT) — a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate — Brain Station 23 (Mar 2025)
- PHP (Laravel) — EDGE Project (Feb 2025)
- Experience Certificate — Code Studio (Apr 2024)
- 4th Place — RUET CodeSmash, team competitive programming contest (Mar 2023)
- 4th Place — RUET CodeSpark, individual competitive programming contest (Oct 2022)